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Dear Editor-in-Chief,

Please consider the enclosed manuscript, *Identifiability and Confounding in Test-Adequacy Metric Validity Studies: A Cross-Language Empirical Analysis*, for publication in the *Journal of Systems and Software* as a research paper.

The manuscript examines the evidential basis of a standard argument in empirical software engineering: that a test-adequacy metric is incrementally valuable because adding it to a regression improves the cross-validated R^2. Rather than proposing a new metric, we ask whether that argument can be statistically resolved at the sample sizes at which it is routinely made. Using Defects4J (Java) and BugsInPy (Python), we show that the increments being claimed are of the same order as their own uncertainty; that their sign depends on cross-validation and resampling choices that are rarely reported; and that project-level confounding alone can manufacture a material increment in up to 82% of runs, a rate that project fixed effects hold below 1%.

Our contribution is methodological rather than a benchmark result, and we believe it speaks to a broad part of the journal's readership: the same incremental-prediction design is used to validate metrics, defect predictors, and effort models across software engineering. We close with a five-check protocol that requires no test executions and can be applied to any published increment-in-R^2 claim, including those in already-published work.

All scripts, collected measurements, and analysis outputs are publicly archived (DOI: 10.5281/zenodo.23432059) and the full pipeline is available at `https://github.com/JunjieChen0/test-adequacy-resolutio`
Every number in the manuscript is produced by the shipped analysis code from the shipped measurements; none is entered by hand.

The manuscript is original, has not been published previously, and is not under consideration by another journal. As the sole author, I have approved the submission, declare no competing interests, and take full responsibility for the content. A declaration of generative AI and AI-assisted technologies is included in the manuscript.

Thank you for considering this submission.

Sincerely,
Junjie Chen